

NEWSLETTER



UTHM DATA CENTER Digital Visit



Summary

On 3 December 2025, the Year 3 Data Engineering students from the Faculty of Computing at UTM visited the Centre for Information Technology (PTM) at UTHM for an academic session focused on the university's digital transformation efforts. The visit was highly relevant to the Enterprise and Design Modelling course, detailing UTHM's strategic ICT journey—from the early KUiTTHO plan to the current Pelan Strategik ICT (2020-2024)—and emphasizing the critical adoption of Enterprise Architecture (EA) as a non-negotiable component of future development. A key part of the session included a tour of the new, partially operational Tier 3 Data Centre (DC). The students gained insight into infrastructure management, noting the use of DCIM, the energy-efficient Hot Aisle/Cold Aisle cooling system, and the robust power redundancy strategy. They also learned about the security measures, including firewalls and Waf, protecting against frequent cyber attacks, and the operational challenges involved in data migration (TCIS to browser-based) and the mandatory institutional usage of the Learning Management System (LMS).

Technologies

During the visit, we explored the technological infrastructure of UTHM's Tier 3 Data Center, which employs a Hot and Cold Aisle containment system and air-cooled condensers to optimize thermal efficiency. The facility is managed using DCIM (Data Center Infrastructure Management) tools and ensures high availability through redundant power sources (PEA & PEB) backed by UPS and generators. For data security and sovereignty, the university utilizes a robust double-firewall architecture and Web Application Firewalls (WAF) to filter international traffic, while relying on on-premise servers replicated at the Government Private Data Centre (PDSA) in Pagoh instead of public clouds.

Issues Discussed

Beyond the physical infrastructure, the session focused on significant operational and cultural challenges, particularly the "culture-first" mindset required for successful digital transformation, as well as common user experience challenges such as Single Sign-On (SSO) lockouts and frequent WiFi authentication failures. Technical discussions revealed that the university does not have a single master database, instead relying on API integrations to connect the various Student Information and Learning Management Systems (LMS).

Reflections

Nurul Adriana Binti Kamal Jefri
(A23CS0258)

Visiting UTHM really helped me see the bigger picture of how IT works in the real world. While touring the Tier 3 Data Center and seeing the hot and cold aisle cooling was technically impressive, the most valuable lesson for me was actually about Enterprise Architecture (EA). I realized that having advanced hardware isn't enough; you need a solid blueprint like TOGAF to make sure all those systems actually work together and support the university's goals. This visit changed my perspective—digital transformation isn't just about the physical "infra" like cables and servers, but equally about the "info" and strategic planning that guides it.

Safiya Nursyahadah
(A23CS0176)

The UTHM visit connected our studies to reality, demonstrating how Enterprise Architecture (EA) frames the university's digital strategy. I gained crucial insight into the Tier 3 Data Centre's design, particularly the power efficiency of the Hot Aisle/Cold Aisle system and essential power redundancy. This experience highlighted the operational importance of meticulous planning, strong cybersecurity, and managing challenges like hardware and software for continuous service.

Iman Abadi
(A23CS0084)

This industrial visit has been an eye-opening experience for me. It showed me that data engineering goes much more than just code to include physical infrastructure, strategy, and human behavior. Most importantly, the realization that "culture comes before technology" taught me that successful digital transformation involves more than just deploying hardware, such as various firewalls or servers, but also overcoming complex user acceptance difficulties, managing technical debt during long-term migrations, and enforcing strict governance to ensure systems remain secure and sustainable.